[Hands-on activity: Use subqueries](#)

Activity Overview

As a data analyst, you will sometimes want to analyze a small subset of data that is contained within a much larger dataset. This is when subqueries can be really useful. As you've been learning, subqueries are queries that are nested inside of another query. Some queries can have many subqueries, while others may have only one. It's important to know how subqueries function and to understand the different components you can use to create them.

In this activity, you will work with queries and subqueries to examine a public health dataset. You will create subqueries to discover which industries receive an unusual number of complaints and, more importantly, which industries are connected to serious health issues. This will allow you to allocate your resources more effectively to safeguard public health.

Review the following scenario. Then complete the step-by-step instructions.

In this scenario, you are a junior data analyst for a multinational food and beverage manufacturer. You and your team are responsible for maintaining the safety of a wide array of food products. Because of the overwhelming number of products on the market, you have been asked to prioritize which products need to be reviewed by your stakeholders.

While it's useful to know which food industries receive the most complaints, the more critical aspect to consider is identifying the complaints that lead to severe health consequences, such as hospital visits.

To complete this task, you'll analyze food event reports for targeted health interventions.

Follow the instructions to complete each step of the activity. Then answer the question at the end of the activity before going to the next course item.

The first step is to access the public dataset. For this activity, you will use a BigQuery public dataset titled **fda_food**, which contains details about food recalls, consumer reactions, product descriptions, and product distribution and quantities.

1. Log in to BigQuery and open the BigQuery Console.

Note: BigQuery frequently updates its user interface. The latest changes may not be reflected in the screenshots presented in this activity, but the principles remain the same. Adapting to changes in software updates is an essential skill for data analysts, and it's helpful for you to practice troubleshooting. You can also reach out to your community of learners on the discussion forum for help.

2. Before you can load the **fda_food** dataset, you need to have **bigquery-public-data** pinned in the Explorer menu of your SQL Workspace. Follow these steps to pin the dataset:

a. Enter **public** in the Explorer menu search box, and press the **Enter**/**Return** on your keyboard.

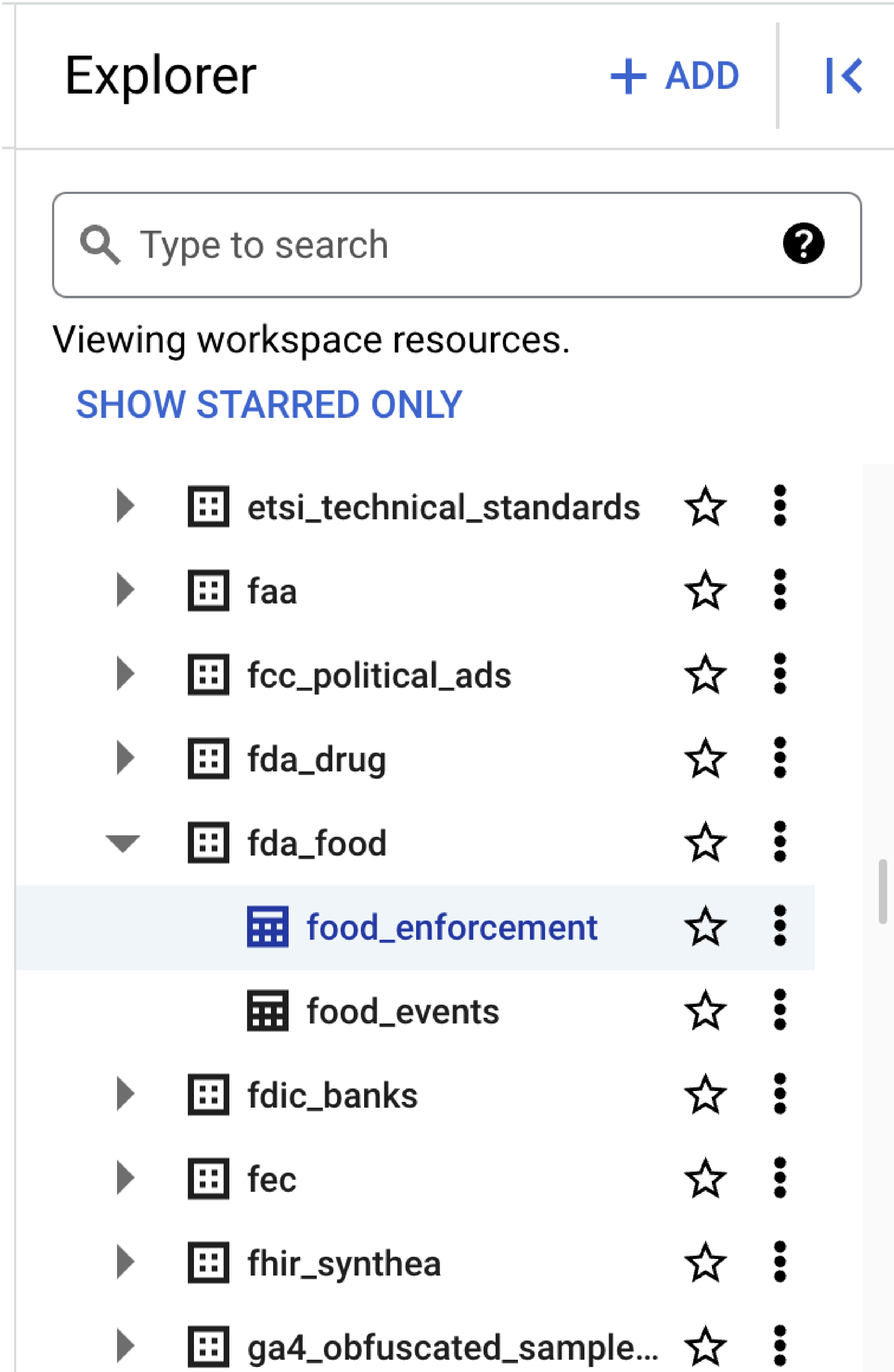
b. Select **SEARCH ALL PROJECTS**.

c. Scroll to find **bigquery-public-data** and select the **star** to pin it.

3. Add the FDA food dataset. Follow these steps:

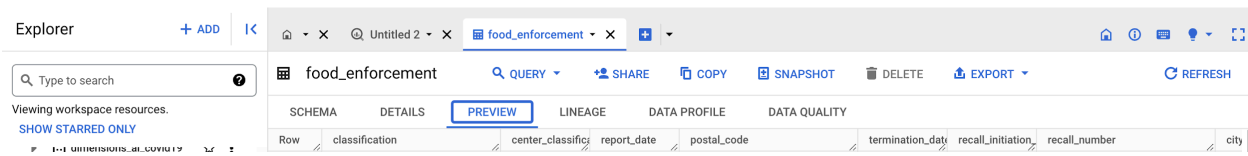
a. Select the **arrow** next to the bigquery-public-data to expand its contents.

b. Scroll to find **fda_food**. You may have to select **SHOW MORE** to view the dataset listed.


Explorer
+ ADD
Type to search
Viewing workspace resources.
SHOW STARRED ONLY
etsi_technical_standards
faa
fcc_political_ads
fda_drug
fda_food
food_enforcement
food_events
fdic_banks
fec
fhir_synthea
ga4_obfuscated_sample...

c. Expand **fda_food** and select **food_enforcement**.

d. Select the **Preview** tab in the viewer to examine the table data.



4. Next, select the **food_events** table from the **fda_food** dataset, then select **Preview** to examine the table data.

Once you have taken some time to explore the tables within the fda_food dataset, you are ready to begin working with subqueries!

Note: Many of the public databases on BigQuery are living records and, as such, are periodically updated with new data. Throughout this course , if your results differ from those you encounter in videos or screenshots, there's a good chance it is due to a data refresh.

To get an initial overview, run a SQL query to identify which food industries have the highest number of complaints. This will provide an initial dataset to work from. To find the top industries for recalls, perform the following steps:

1. Create a new query.

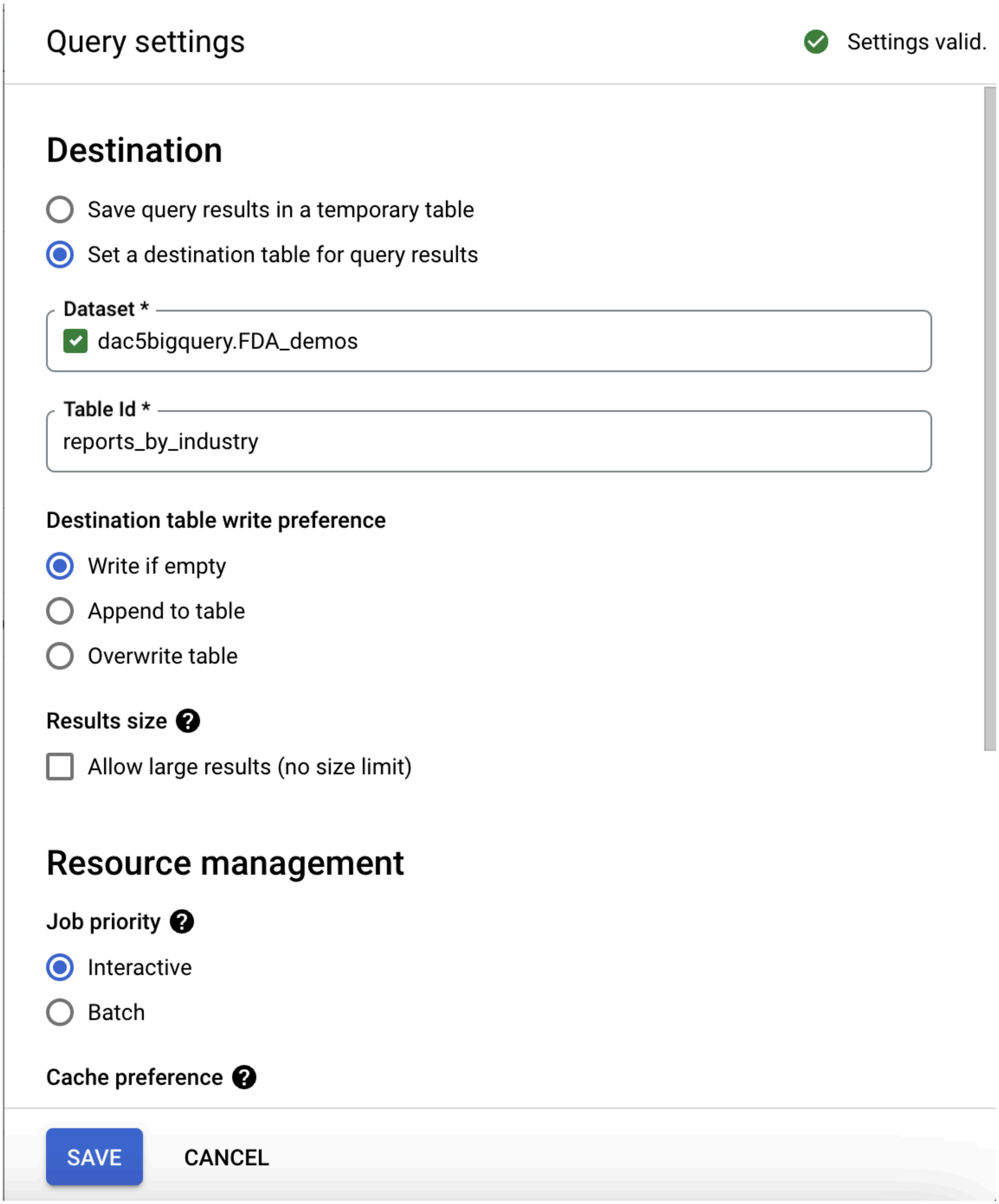
2. Enter the following in the query editor:

```
SELECT
products_industry_name,
COUNT(report_number) AS count_reports
--SELECT is used to identify the product industries by name. COUNT will
count the number of reports and label them as count_reports.
FROM bigquery-public-data.fda_food.food_events
GROUP BY products_industry_name
ORDER BY count_reports DESC
LIMIT 10;
--The query will group the product industries by name so the report
results will fall under each industry title. ORDER BY and DESC will tell
the query to order the output by count_reports in descending order so, the
industry with the most amount of reports will appear at the top of the
output table. LIMIT will limit the results to ten industries with the
highest numbers of reports.
```

Note: Depending on the type of web browser you are using, you may have issues conducting a direct copy + paste of the above syntax directly from this assignment to your BigQuery workspace. If you are experiencing this issue, open two separate windows on your screen to manually enter the above information into the BigQuery workspace area.

3. You have the option to save your query as a table. To do this, first create a new dataset named **demos**. Select the **MORE** menu from the Query Editor and open the Query Settings menu. In the Query Settings menu, select **Set a destination table** for query results. Click in the **Dataset** field and select **Create New Dataset**. A new pop-menu will open. In the **Dataset Id** field, enter **demos**. Click **Create Dataset.** Now, you'll return to the Query Settings menu, and the **Dataset** field should automatically populate with your project Id followed by **demos.** In the **Table Id** field, enter the name **reports_by_industry**. Click **Save**.

Note: In the **Dataset** field, **demos** should be preceded by your unique project id. Do not enter the Id shown in the screenshot.


Query settings
Settings valid.
Destination
Save query results in a temporary table
Set a destination table for query results
Dataset *
dac5bigquery.FDA_demos
Table Id *
reports_by_industry
Destination table write preference
Write if empty
Append to table
Overwrite table
Results size
Allow large results (no size limit)
Resource management
Job priority
Interactive
Batch
Cache preference
SAVE
CANCEL

4. Select **RUN**. The query will save as a new table in your demos dataset.

The **Query results** window will populate with a table containing two columns and 10 rows. The first column is `products_industry_name`, and the second is `count_reports`. The 10 different industries are listed in descending order in the rows.

Take some time to check the different industries and the differences in report numbers. You can modify your search to examine the different symptoms that consumers experienced (`reactions`) based on products (`products_brand_name`) as well as the `outcomes`, or consequences, from each food event.

Query results   SAVE RESULTS   EXPLORE DATA

JOB INFORMATION   RESULTS   CHART PREVIEW   JSON   EXECU

Row	products_industry_name	count_reports
1	Vit/Min/Prot/Unconv Diet(Human/Animal)	83516
2	Cosmetics	76363
3	Nuts/Edible Seed	5413
4	Vegetables/Vegetable Products	4337
5	Soft Drink/Water	3294
6	Bakery Prod/Dough/Mix/Icing	3262
7	Fruit/Fruit Prod	2996
8	Fishery/Seafood Prod	2808
9	Cereal Prep/Breakfast Food	2267
10	Dietary Conventional Foods/Meal Replacements	2203

With a list of the industries receiving the most complaints, your next step is to find out which of these complaints led to hospitalizations. Filter the initial list accordingly and focus solely on complaints that resulted in hospital visits. To do this:

1. Create a new query.

2. Enter the following into the editor:

```
SELECT
products_industry_name,
COUNT(report_number) AS count_hospitalizations
```

```
FROM
bigquery-public-data.fda_food.food_events
WHERE products_industry_name IN
(SELECT
products_industry_name
FROM
bigquery-public-data.fda_food.food_events
GROUP BY products_industry_name
ORDER BY COUNT(report_number) DESC LIMIT 10)
AND outcomes LIKE '%Hospitalization%'
--The AND operator displays a record if all the conditions are TRUE.
--The LIKE operator is used in a WHERE clause to search for a specified
pattern in a column.
GROUP BY products_industry_name
ORDER BY count_hospitalizations DESC;
```

The outer query used here is just like the previous query. `SELECT` calls the product industry names, and `COUNT` counts the number of reports and labels them as count_hospitalizations. The query is told to pull this data from the `food_events` table.

The subquery begins with a `WHERE` clause and an `IN` operator—these will allow the subquery to specify multiple values. The second `SELECT` statement tells the subquery to pull the product industry names `FROM` the `food_events` table in the `FDA_food` dataset. Results are then grouped by product industry name and ordered in descending order. `LIMIT` tells the query to limit the number of results to 10. `AND` and `LIKE` are used to indicate if hospitalizations are present within the `outcomes` field.

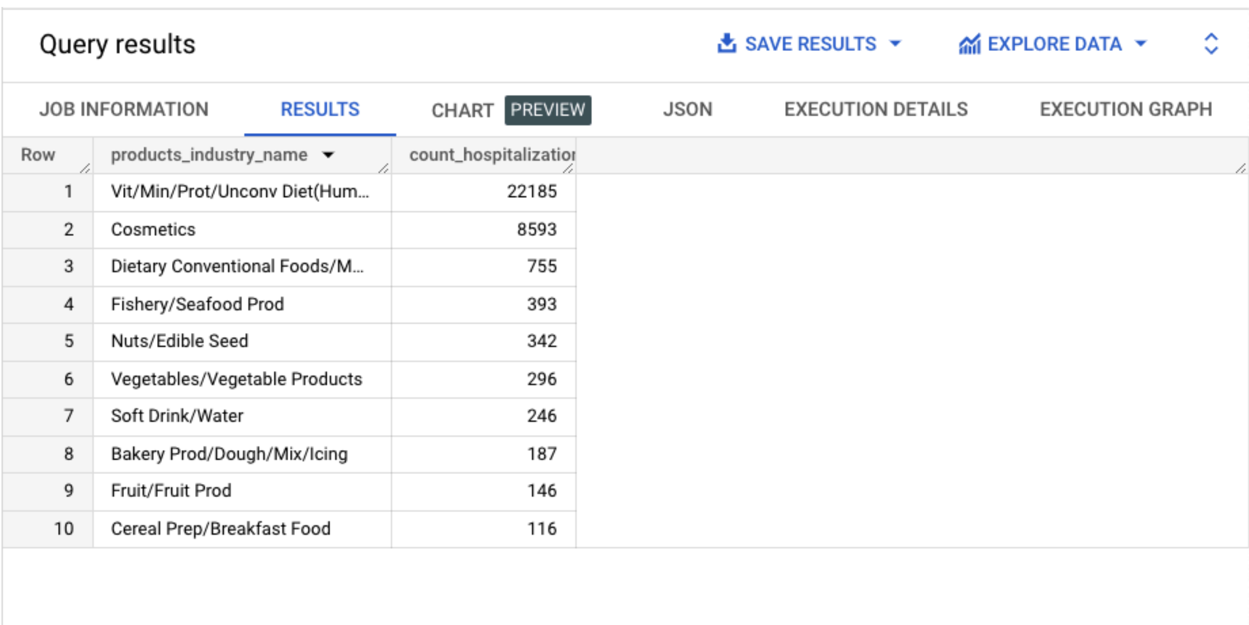
The outer query is then closed by grouping results by `product_industry_name` and ordered in descending order by the number of hospitalizations recorded.

3. You can also save this query as a table. Before running the query, select the **MORE** menu from the Query Editor and open the Query Settings menu. In the Query Settings menu, select **Set a destination table** for query results. Set the dataset option to **demos** and name the table **hospitalizations_by_industry**.

4. Select **RUN**. The query will save as a new table in your demos dataset.

The **Query results** window will populate with a table containing two columns and ten rows. The first column is `products_industry_name`, and the second is `count_hospitalizations`. The 10 different industries are listed in descending order in the rows.

Take some time to review the different industries and the differences in reported hospitalizations. You can modify your search to examine the different symptoms that consumers experienced (`reactions`) based on products (`products_brand_name`); the dates in which the food events occurred (`date_started`); and the `outcomes`, or consequences, from each food event. These modifications will allow you to find and analyze any links among fields within the datasets for your research.



Row	products_industry_name	count_hospitalizatior
1	Vit/Min/Prot/Unconv Diet(Hum...	22185
2	Cosmetics	8593
3	Dietary Conventional Foods/M...	755
4	Fishery/Seafood Prod	393
5	Nuts/Edible Seed	342
6	Vegetables/Vegetable Products	296
7	Soft Drink/Water	246
8	Bakery Prod/Dough/Mix/Icing	187
9	Fruit/Fruit Prod	146
10	Cereal Prep/Breakfast Food	116

Great work! Through this process, you discovered which industries receive an unusual number of complaints and, more importantly, which are connected to serious health issues. This knowledge allows you to allocate your resources more effectively to safeguard public health. By completing this activity, you have created subqueries using `SELECT` statements with `FROM, GROUP BY, WHERE, IN, AS, COUNT`, and `ORDER BY` clauses, as well as `AND` and `LIKE` operators to search and analyze the query results relating to reports of food events by industry.

1. **Reflection** 1 / 1 point

In the text box below, write 2-3 sentences (40-60 words) in response to each of the following questions:

- Why is working with subqueries an important skill for data analysts to learn?
- What are some benefits of using subqueries?

Working with subqueries is an important skill for data analysts to learn because if you want to filter a lot of things from a single query and also want to do many operations like sum, count, and avg, etc., then a subquery is ideal for that because after writing the whole query, the result is generated very smoothly, and you check all your desired answers from that query.

That's why it's an important skill. In a single query you can find a lot of answers. And you may structure the query in a very good way such that you can find those answers very easily and get the good exposure using subqueries.

Feedback
Congratulations on completing this hands-on activity! In this activity, you created queries and subqueries to analyze sections of a large dataset. An effective response might include:

- Learning how to create queries and subqueries is a key component to the ability to research and deliver useful findings to your leadership or businesses.
- Subqueries help you identify and work with smaller sections of data from a large dataset.

Subqueries can be challenging—they take practice and repetition to execute well. Now that you have a starting point, take some time to practice working with different datasets within the bigquery-public-data database and building out your own subqueries.